Fórmulas

$F = k\cdot|q_1|\cdot|q_2| \,/\, d^2$ | $E = k\cdot|Q| \,/\, d^2$

$Fx = F\cdot\cos\theta$ $\quad Fy = F\cdot\operatorname{sen}\theta$ | $F_R = \sqrt{(\Sigma Fx^2 + \Sigma Fy^2)}$

$\beta = \tan^{-1}(|\Sigma Fy| \,/\, |\Sigma Fx|)$ | $d = \sqrt{(k\cdot q_1\cdot q_2 \,/\, F)}$

$q_2 = F\cdot d^2 \,/\, (k\cdot q_1)$ | $k = 9\times10^9\ N\cdot m^2/C^2$

$\mu = 10^{-6}\cdot n = 10^{-9}\cdot m = 10^{-3}\cdot p = 10^{-12}\cdot cm \div 100 = m$
θ: I = β · II = 180−β · III = 180+β · IV = 360−β
Igual signo **repele** · distinto **atrae** · E **sale** de + y **entra** a −
Calculadora en grados: `60` `cos` = 0.5 ✓

Ej 1 · Fuerza resultante

componentes

Fx = −10 N, Fy = 14 N. Hallar F_R y θ.

✍ A MANO	📱 CALCULADORA
$F_R = \sqrt{((-10)^2 + 14^2)}$	`10` `x²` →100 · `14` `x²` →196
$F_R = \sqrt{296}$	`100` `+` `196` `=` `√x` →17.20
F_R = 17.20 N	
$\gamma = \tan^{-1}(14/10) = \tan^{-1}(1.4)$	`14` `÷` `10` `=` `2nd` `tan⁻¹` →54.46
Cuadrante II: $\theta = 180 - 54.46$	`180` `−` `54.46` `=` →125.54
θ = 125.54° (profe ≈125°)	

Ej 2 · Coulomb en línea

F_R sobre q_1

q_1 = 4µC, q_2 = 2µC, q_3 = 1µC (todas +). q_1 —2 m— q_2 —2 m— q_3

✍ A MANO	📱 CALCULADORA
$F_{12} = 9\times10^9\cdot2\times10^{-6}\cdot4\times10^{-6} \,/\, 2^2$ $10^{9-6-6} = 10^{-3}$	`9` `×` `2` `×` `4` →72 `72` `÷` `4` →18
F_{12} = 18×10⁻³ N	
$F_{13} = 9\times10^9\cdot4\times10^{-6}\cdot1\times10^{-6} \,/\, 4^2$	`9` `×` `4` →36 · `36` `÷` `16` →2.25
F_{13} = 2.25×10⁻³ N	
Profe: $F_R = F_{13} - F_{12}$	`2.25` `−` `18` `=` →−15.75
F_R = 15.75×10⁻³ N	(pizarra)

⚠ Todas +: las dos empujan q_1 a la izquierda y se **suman**: 18 + 2.25 = **20.25×10⁻³ N** ←

Ej 3 · Coulomb en triángulo

F_R sobre q_1

q_1 = +7µC (arriba), q_3 = +2µC (abajo izq.), q_2 = −4µC (abajo der.), L = 0.5 m, 60°

✍ A MANO	📱 CALCULADORA
$F_{13} = 9\cdot7\cdot2\times10^{-3} \,/\, 0.5^2$	`9` `×` `7` `×` `2` →126 · `126` `÷` `0.25` →504
$F_{12} = 9\cdot7\cdot4\times10^{-3} \,/\, 0.5^2$	`9` `×` `7` `×` `4` →252 · `252` `÷` `0.25` →1008
F_{13} = 0.504 N · F_{12} = 1.008 N	
$\Sigma Fx = 0.504\cdot\cos60 + 1.008\cdot\cos60 = 0.252 + 0.504 = 0.756$	`60` `cos` →0.5 `0.504` `×` `0.5` · `1.008` `×` `0.5`
$\Sigma Fy = 0.504\cdot\operatorname{sen}60 + 1.008\cdot\operatorname{sen}60 = 0.436 + 0.873 = 1.309$	`60` `sin` →0.866 `0.504` `×` `0.866` · `1.008` `×` `0.866`
$F_R = \sqrt{(0.756^2 + 1.309^2)}$	`0.756` `x²` `+` `1.309` `x²` `=` `√x` →1.512
$\beta = \tan^{-1}(1.309/0.756)$	`1.309` `÷` `0.756` `=` `2nd` `tan⁻¹` →60
F_R ≈ 1.51 N · β ≈ 60°	(pizarra: 1.53 N · 59.31°)

⚠ q_2 es − y atrae q_1 hacia abajo: $F_{12}y$ = −0.873 → ΣFy = −0.437 → **F_R = 0.873 N a −30°**

Ej 4 · Campo eléctrico en línea

E_R en P

q_1 = +10µC —2 m— P —3 m— q_2 = +8µC

✍ A MANO	📱 CALCULADORA
$E_1 = 9\times10^9\cdot10\times10^{-6} \,/\, 2^2$ $10^{9-6} = 10^3$	`9` `×` `10` →90 · `90` `÷` `4` →22.5
E_1 = 22.50×10³ N/C →	(sale de q_1 +)
$E_2 = 9\times10^9\cdot8\times10^{-6} \,/\, 3^2$	`9` `×` `8` →72 · `72` `÷` `9` →8
E_2 = 8×10³ N/C ←	(sale de q_2 +)
Opuestos: $E_R = (22.50 - 8)\times10^3$	`22.5` `−` `8` `=` →14.5
E_R = 14.50×10³ N/C →	

Ej 5 · Campo en triángulo

E_R y θ en P

q_1 = −8nC (abajo izq.), q_2 = +6nC (abajo der.), L = 0.2 m, P arriba.
E_1 va hacia q_1 → 240° · E_2 sale de q_2 → 120°

✍ A MANO	📱 CALCULADORA
$E_1 = 9\cdot8 \,/\, 0.2^2$ $(10^{9-9} = 1)$ $E_2 = 9\cdot6 \,/\, 0.2^2$	`72` `÷` `0.04` →1800 `54` `÷` `0.04` →1350
$E_1x = 1800\cdot\cos240 = -900$ $E_2x = 1350\cdot\cos120 = -675$	`240` `cos` `×` `1800` `=` `120` `cos` `×` `1350` `=`
$E_1y = 1800\cdot\operatorname{sen}240 = -1558.85$ $E_2y = 1350\cdot\operatorname{sen}120 = 1169.13$	`240` `sin` `×` `1800` `=` `120` `sin` `×` `1350` `=`
$\Sigma Ex = -1575$ · $\Sigma Ey = -389.71$	suma cada columna
$E_R = \sqrt{(1575^2 + 389.71^2)}$	`1575` `x²` `+` `389.71` `x²` `=` `√x` →1622.50
$\beta = \tan^{-1}(389.71/1575)$ III: $\theta = 180 + \beta$	`389.71` `÷` `1575` `=` `2nd` `tan⁻¹` →13.90 `180` `+` `13.90` →193.90
E_R ≈ 1622.50 N/C · θ ≈ 193.90°	(pizarra: 1622.32 · 193.87°)

Ej 6 · Despejar la distancia

cuaderno

$q_1 = 4\mu C$, $q_2 = 2\mu C$, $F = 0.2$ N. Hallar d.

✍️ A MANO	📱 CALCULADORA
$d = \sqrt{(k \cdot q_1 \cdot q_2 \ / \ F)}$	
$9\times10^{9} \cdot 4\times10^{-6} \cdot 2\times10^{-6} = 72\times10^{-3} = 0.072$	9 × 4 × 2 →72 (coma 3 lugares → 0.072)
$d = \sqrt{(0.072 \ / \ 0.2)} = \sqrt{0.36}$	0.072 ÷ 0.2 = √x →0.6
d = 0.6 m	

Ej 7 · Dos cargas iguales

1er parcial

d = 200 cm, $F = 9.5\times10^{-14}$ N, $q_1 = q_2 = q$. Hallar q.

✍️ A MANO	📱 CALCULADORA
$d = 200 \div 100 = 2$ m $q^2 = F \cdot d^2 \ / \ k$	200 ÷ 100 →2
$q^2 = 9.5\times10^{-14} \cdot 4 \ / \ 9\times10^{9} = 4.22\times10^{-23}$	9.5 × 4 →38 · 38 ÷ 9 →4.222 exponente: −14 − 9 = −23
exponente par: 42.2×10^{-24} $q = \sqrt{42.2} \times 10^{-12}$	42.22 √x →6.498
$q \approx 6.5\times10^{-12}$ C (cada una)	